

Multiple Choice (10 marks)

1. Which body composition method would you use to measure body mineral content?
 - (a) Magnetic Resonance Imaging
 - (b) Dual-energy X-ray absorptiometry
 - (c) Isotope dilution
 - (d) Skinfold Thickness
2. How much dietary protein should athletes consume daily?
 - (a) <0.8 g protein per kg bodyweight per day
 - (b) 0.8-1.2 g protein per kg bodyweight per day
 - (c) 1.2-1.7 g protein per kg bodyweight per day
 - (d) >2.0 g protein per kg bodyweight per day
3. Higher dietary sodium (salt) intake is generally associated with:
 - (a) Decreased calcium excretion in the urine
 - (b) Increased risk of fracture
 - (c) Decreased dietary calcium absorption
 - (d) Negative calcium balance and bone mineral loss
4. Nutrients may pose health risks:
 - (a) When intakes exceed established ULs (Upper Levels)
 - (b) At any level greater than a recommended dietary intake
 - (c) Only when intakes are less than those required for nutritional benefits
 - (d) Under no known conditions of usual intake
5. The effects of iron on certain aspects of socio-emotional development (shyness, orientation/engagement and response to unfamiliar pictures) is due to:
 - (a) The role of iron in myelination
 - (b) The role of iron in neurochemistry
 - (c) The role of iron in dopamine production
 - (d) None of the above

True / False (10 marks)

Answer True or False

6. True or False: Elimination of toxins is the most important function of the gastrointestinal tract right after eating.
7. True or False: The absorption of glucose in the gastrointestinal tract is linked to the uptake of sodium.
8. True or False: High protein diets are ineffective for weight loss because they do not influence appetite or body composition.
9. True or False: Type IIb muscle fibers have low oxidative capacity and are not known for high fatigue resistance.
10. True or False: Bariatric surgery can reduce the incidence of type 2 diabetes in obese individuals by 80-90%.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the characteristics of the Atherogenic Lipoprotein Phenotype (ALP) and explain how these lipoprotein abnormalities can impact overall health.
12. Discuss the nutritional status of older adults, focusing on the risks associated with iron stores, and explain why high iron stores may pose a greater risk than iron deficiency in this population.

End of Paper